Transmembrane Potential: Generation and Propagation of Neuronal, Muscle, and Cardiac Action Potentials, and the Pacemaker and Pacemaker Potential

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Introduction

Electricity and magnetism are two of the most fundamental physical phenomena underlying biological systems. In humans and other living things, electricity informs the inner workings of the transmembrane potential, which plays a key role in cell-to-cell communication.

Transmembrane potential, sometimes referred to as 'membrane potential' or 'membrane voltage', describes the difference between the internal and external faces of a biological membrane—a cellular feature serving as a diffusion barrier to ion movement. For cells, electric potential is established due to differences in ion concentration on either side of its semipermeable membrane. Typical values of the transmembrane potential range from -40 to -70 millivolts.

This paper will dig deeper into the transmembrane potential and its biological implications, specifically in the context of the generation and propagation of neuronal, skeletal muscular, and cardiac action potentials, and pacemaker potentials. An important application of this concept, the pacemaker, will also be looked at closely in terms of its mechanism and significance in research.

Neuronal Action Potential

The nervous system's function in the body is to receive information from the internal and external environment and integrate it via a response loop [1]. For this to occur, neurons must generate what is called an action potential to signal our body and trigger a response [1]. Thus, neurons play a crucial role in transmitting information and signals over long distances electrically [1].

The cell membrane acts as an electrical insulator, meaning ions are not free to move around the cell membrane[1]. Meaning, we will have ions found in the inside of the cell in the intracellular fluid (ICF) and outside the cell in the extracellular fluid (ECF). Cell membranes will contain voltage-gated ion channels which allows for the selective flow of ions across the cell membrane depending on the concentration gradient of the ions (due principles of osmosis) [1]. Since these channels are voltage-gated, the membrane will also have a potential depending on the ions found in the ICF and ECF[1]. We can find the equilibrium potential of a specific ion across the membrane using the following Nernst's equation (Figure 1):

$$E_{ion} \text{ (in mV)} = \frac{61}{z} \log \frac{[\text{ion}]_{out}}{[\text{ion}]_{in}}$$

Equation 1. *Nernst's equation - where 61=2.303*RT/F, E_{ion}=equilibrium potential for a specific ion, z=electrical charge of ion[1]*

At rest, the cell membrane potential for neurons is -70mV to -90mV; this is typically more negative due to excessive negative charges found in the ICF[1]. Now with a solid understanding of the electrical nature of neurons, let's get into the generation and propagation of action potential through neurons. In the presence of a stimulus neurons will be signalled to open or close their ion channels[1]. This will result in a change in membrane potential as ions move into or out the cell, creating waves of depolarization (voltage becomes less negative) and hyperpolarization (voltage becomes more negative)[1]. When a neuron reaches a minimum depolarization of approximately -55mV via the opening and closing of ion channels, it will trigger an action potential which will undergo a series of phases (Figure 1)[1].

The action potentials will propagate from the cell body to the axon terminal of a neuron in only one direction due to the refractory period. The refractory period occurs after an action potential has been generated in an area, this leads to the depolarization of an adjacent area where another action potential can be triggered when threshold is reached[1]. The action potentials will travel down the axon from positive active areas to inactive regions. Back flow will be prevented as regions from previous action potentials are undergoing refractory periods and cannot trigger new action potentials (Figure 2) [1].

Once these action potentials reach the axon terminal they can ultimately result in cell-to-cell communication via synapses. Where a neuron will transmit these signals from the presynaptic axon terminal to a postsynaptic cell (another neuron or target cell) across the synaptic cleft (the space between the axon terminal and the cell) or via gap junctions [1]. This signaling can occur electrically (using signaling and gap junctions) or chemically (using neurotransmitters) which will bind to receptors on the postsynaptic cell and result in a response at the target cell [1].

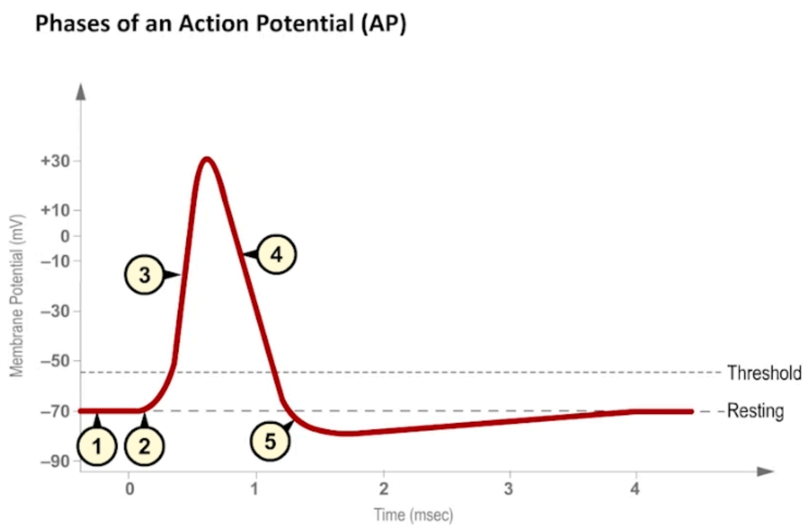


Figure 1. Phases of an Action Potential
(1,2) Initial depolarization due to stimulus: Stimulus initiates a depolarization that passes the threshold and opens voltage Na+ channels
(3) Rising phase: rapid depolarization occurs due to Na+ channels opening and Na+ going into ICF
(4) Falling phase: repolarization (returning to resting membrane potential) due to K+ channels opening and leaving cell and Na+ channels start to close
(5) Hyperpolarization: occurs due to K+ channel remaining open while Na+ channel are closed
(6) Return to resting potential: voltage gated K+ channels close and membrane returns to resting potential[1]

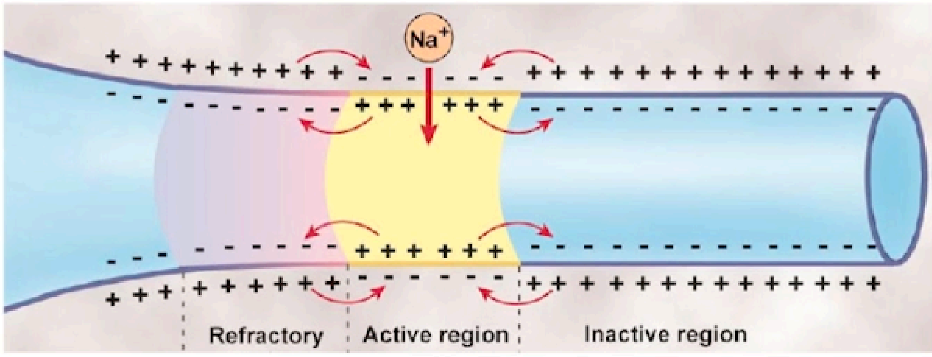


Figure 2. Propagation of action potentials from active regions to inactive regions down the axon[1]

Muscular Action Potential and Muscle Contraction

Muscles (or muscle tissues, muscle fibres) are one of the organs in the human body where electricity and electrical activity play a vital role in its underlying mechanism. Muscles are specialized tissues whose objective is to convert biochemical energy to mechanical energy or work[1].

Muscle contraction begins with an action potential, which travels through motor neurons (Figure 2)[1]. The motor neuron then reaches a muscle cell via the neuromuscular junction. Acetylcholine (Ach), a neurotransmitter, is released and binds to receptors on the muscle fiber and triggers membrane channels open allowing entry of Na+ ions[2]. This sodium influx sends a message allowing the release of stored calcium ions within the muscle fiber. Ca2+ ions diffusing into the muscle fiber changes the relationship between the protein chains within muscle cells, leading to muscle contraction (Figure 3)[3].

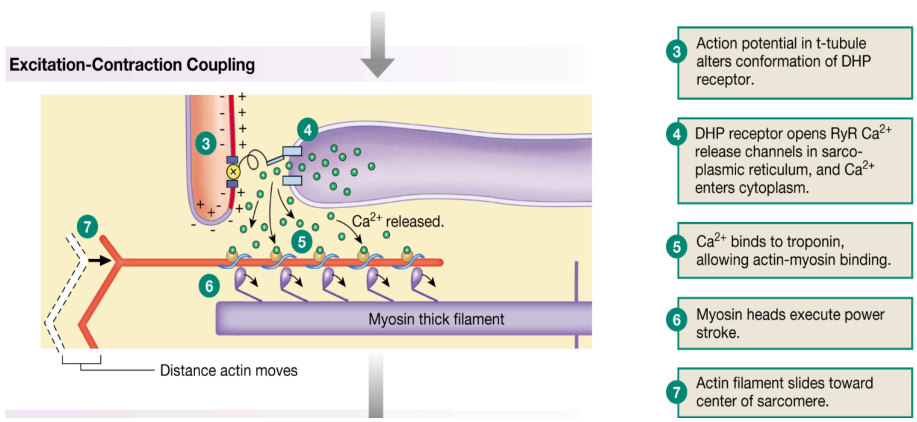


Figure 3. The excitation-contraction coupling, which occurs upon the initiation of a muscle action potential, connects the action potential generated in the sarcolemma, or muscle fibres, and the start of a muscle contraction[3]

Skeletal muscle action potential differs from cardiac and neuronal action potential in terms of the duration. In neurons, the duration is normally 1 ms and in cardiac cells it ranges from 200 to 400 ms while in skeletal muscle cells, the cycle usually lasts for 2 to 5 milliseconds[4]. One method used to measure the electrical activity of a muscle or muscle fiber is through electromyography. This technique is performed by placing electrodes, which act like sensors, on the skin over a muscular region. Electromyography plays a big role in detecting neuromuscular abnormalities. Electromyograms show the overall electrical activity or action potential of multiple muscle fibres. Muscle contraction generates multiple simultaneous electric potentials in multiple fibers being activated; hence, the electromyogram displays potential differences plotted against time. Unlike electrocardiography, the EMG does not show a series of waves with predictable patterns but spike-like signals indicating the intensity of muscle activity during contraction[2].

Cardiac Action Potential

In the heart, the signal for muscle contraction does not come from the nervous system, but from the heart itself [1]. There are two types of muscle cells that make up the heart: contractile cells and autorhythmic cells (also known as pacemaker cells). The signal for myocardial contraction comes from autorhythmic cells, which can generate action potentials on their own. From there, the signal spreads to other cells via specialized structures called gap junctions, which allow electrical impulses to pass from cell to cell so that the heart muscle can contract simultaneously. The electrical impulses generated by the autorhythmic cells travel through gap junctions to the contractile cells of the atria, generating an action potential in those cells and causing atrial contraction (Figure 5). Then, the impulses travel through specialized conductive fibers to the bottom of the heart, triggering a wave of depolarization that spreads upward through the contractile cells of the ventricles and causing ventricular contraction [1].

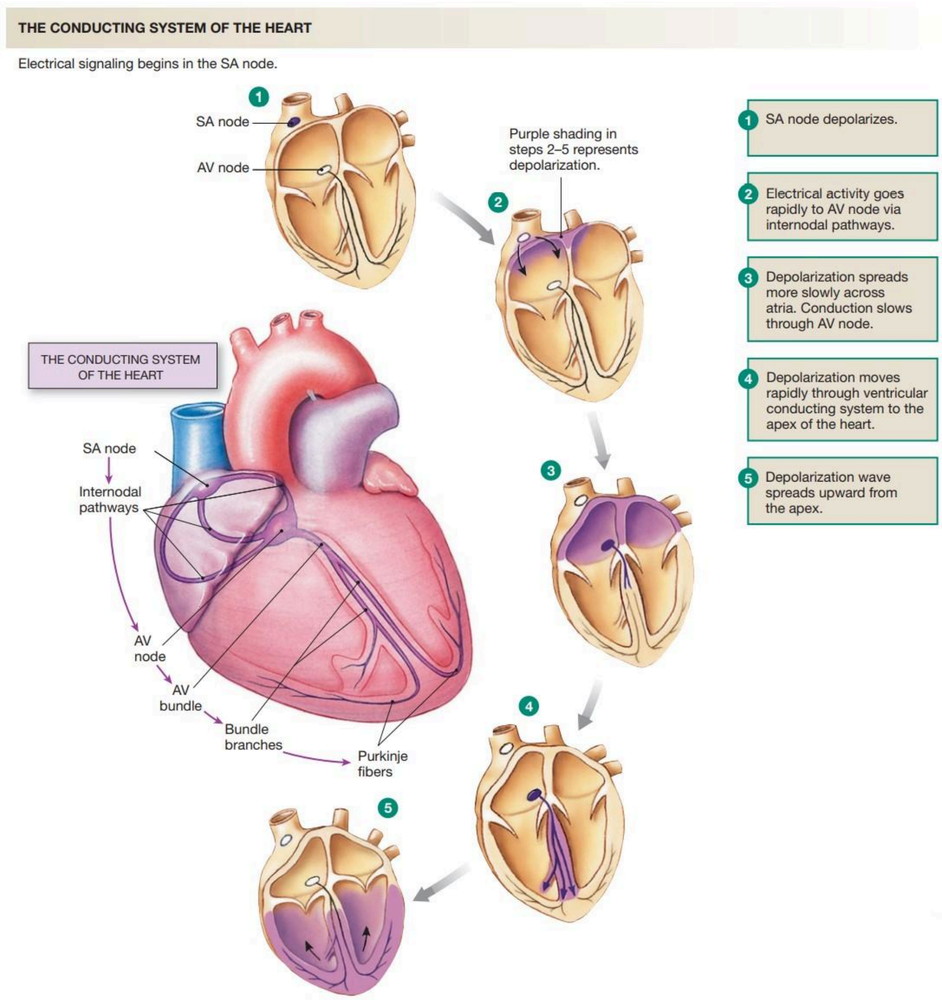


Figure 4. The conducting system of the heart. [1]
(1) Electrical impulses originate from autorhythmic cells in the sinoatrial node.
(2) Depolarization spreads to the AV node.
(3) Depolarization spreads across contractile cells of the atria in a downward path, causing the atria to contract and eject blood into the ventricles.
(4) Depolarization travels down specialized fibers which carry the electrical impulse rapidly to the bottom of the heart.
(5) Depolarization reaches the contractile cells of the ventricles, traveling up the ventricles in an upward fashion. This allows the bottom of the heart to contract first, ejecting blood upwards and out of the heart.

While action potentials can be generated in both autorhythmic and contractile cells, the action potentials generated in either cell type are distinct in several ways [1]. Autorhythmic cells have a unique ability to generate spontaneous action potentials, which can be attributed to their unstable membrane potential. Autorhythmic cells have an unstable membrane potential that starts at -60 mV and slowly drifts upward due to an influx of positively charged ions. Once the membrane potential reaches a certain level, an action potential is automatically triggered. This process is further elucidated in Figure 5 below [1].

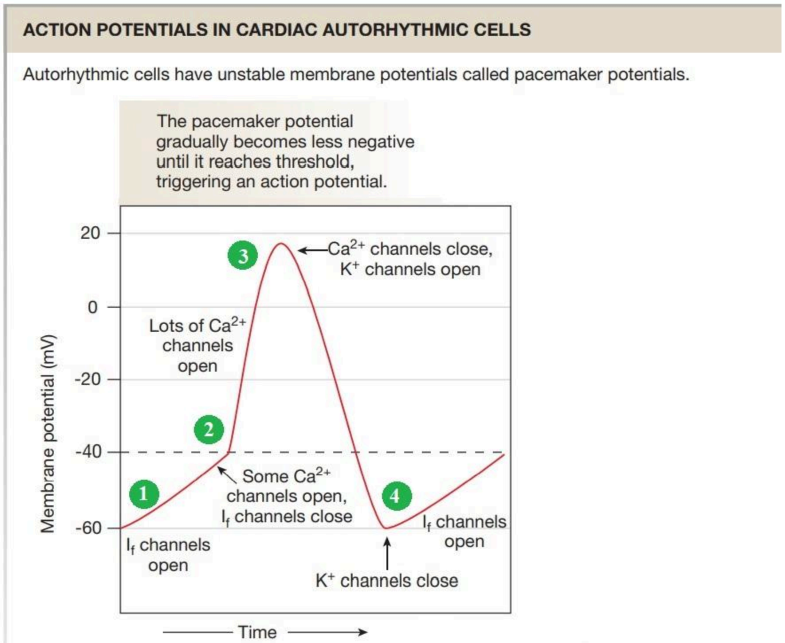
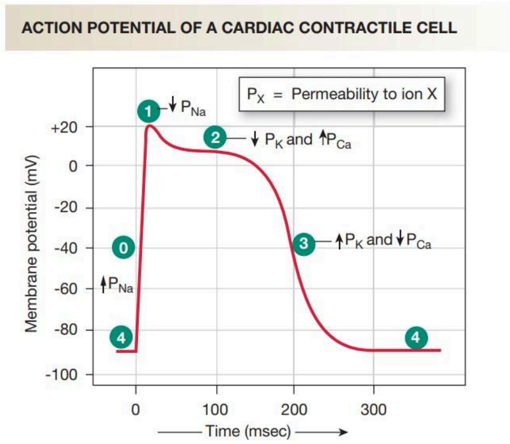


Figure 5. Action potential in cardiac autorhythmic cells. [1]
(1) Slow upward drift of the membrane potential caused by influx of Na^+ ions through "funny" (I_f) channels, depolarizing the membrane. As the action potential threshold of -40 mV approaches, I_f channels close and one set of Ca^{2+} channels open.
(2) Threshold is reached, triggering the opening of a different set of Ca^{2+} channels. Ca^{2+} rapidly rushes into the cell, causing a steep depolarization.
(3) Peak is reached. Ca^{2+} channels close and K^+ channels are completely open, leading to efflux of K^+ ions. This causes the membrane to depolarize.
(4) K^+ channels close and the cycle restarts.

Unlike autorhythmic cells, contractile cells cannot generate spontaneous action potentials, as they have a stable resting membrane potential around -90 mV [1]. An action potential is triggered only when a current of positive ions enters the cell from neighbouring cells via gap junctions. The action potential in contractile cells is explained in more detail in Figure 6. One unique feature of the action potential in contractile cells is the increased length of the action potential, which is 40-200 times longer than that of a typical neuron or skeletal muscle cell. This is caused by the plateau in membrane potential after the peak. The increased length of the action potential allows sufficient time for the cardiac muscles to relax so that the ventricles can fill with blood [1].



Phase*	Membrane channels
0	Na^+ channels open
1	Na^+ channels close
2	Ca^{2+} channels open; fast K^+ channels close
3	Ca^{2+} channels close; slow K^+ channels open
4	Resting potential

*The phase numbers are a convention.

Figure 6. Action potential of cardiac contractile cells. [1]
(4) Resting membrane potential of -90 mV.
(0) When a wave of depolarization arrives from a neighbouring cell through gap junctions, the membrane potential becomes more positive, triggering voltage-gated Na^+ channels to open. The influx of Na^+ rapidly depolarizes the cell membrane.
(1) Na^+ channels have closed. K^+ flows out of the cell through open K^+ channels causing a slight repolarization.
(2) Some K^+ channels have closed, decreasing the K^+ current. However, voltage-gated Ca^{2+} channels are now fully open and Ca^+ enters the cell. The combination of these two events leads to a plateau in the membrane potential.
(3) Ca^{2+} channels close and a different set of K^+ channels open, causing the membrane to begin to repolarize much more rapidly. K^+ exits the cell until resting membrane potential is reached.

An Introduction to Pacemakers

An electronic pacemaker is a medical device that is used to treat certain arrhythmias [5]. It does this by discharging an electrical current within the cardiac chambers and employs one of three functions depending on what is needed for proper cardiac contraction: inhibit output, trigger output or pace in a different chamber after a timed delay [6]. A pacemaker is also able to increase the pacing rate depending on the physical, mental or emotional exertion the client has, for example, this is used to increase exercise capacity where the need for greater pacing is required [6]. Pacemakers nowadays are able to "sense" meaning they are able to determine the timing of the cardiac depolarization where the lead is in [6].

Parts of the Pacemaker

Generally, the pacemaker is made up of two parts; the "pulse generator" and the "leads" [6]. The pulse generator consists of a battery and electronics that control the pacemaker and is programmed by a 4 or-5 letter code that signifies what the pacemaker should do and where to send the signal [6]. The leads can be thought of as wires that connect the pulse generator to the myocardium to deliver the pulse [6]. The leads are made up of two parts, the conductor and the tips [6].

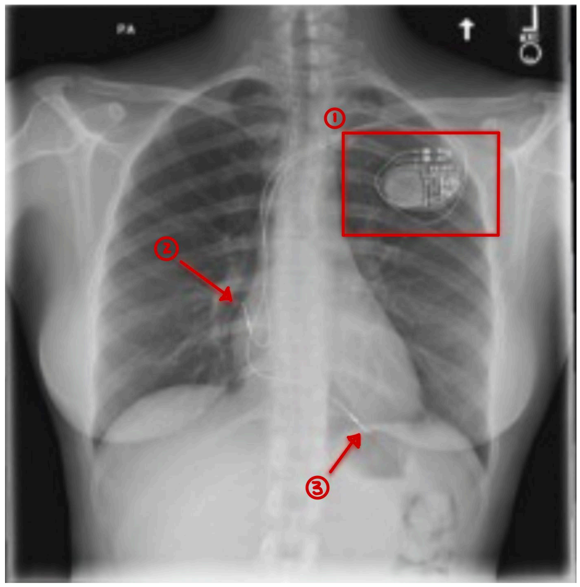


Figure 7. An x-ray of a pacemaker implant where (1) is the pulse generator, (2) and (3) are the leads that attach to the heart with the conducting tips being seen [7]

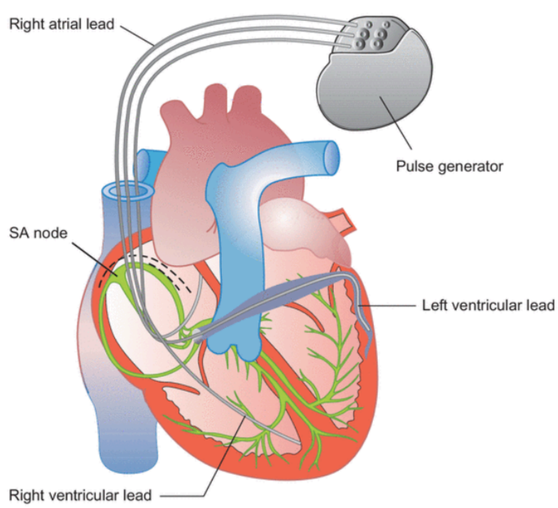


Figure 8. A schematic of the pacemaker, its parts and placement in the heart. [5]

The potential difference between two electrodes is responsible for the pacing to occur. In this way, permanent pacemakers supply a constant voltage [6]. The use of Ohm's Law of V=IR, explains the amount of resistance the pacemaker parts are designed to have [6]. The lead conductors in pacemakers are designed to have lower resistance to allow for the maximal amount of current to pass through the leads which minimizes the energy [6]. The lead tips are designed to have high resistance to lower the amount of current used which then increases the overall battery length per pacing pulse [6].

The amplitude range for the atrial lead is between 1.5-5mV and the range for ventricular leads are 5-25mV. If the amplitude range for these values are under these ranges, it can cause "undersensing" which then will not deliver the proper pace pulse [6]. Pacemakers will emit pulses that have a duration between 0.5 to 25 milliseconds and output 0.1 to 15V depending on what is needed. [8]

Future Avenues of Pacemaker Research

While electronic pacemakers have been a beneficial form of medical technology, research and engineering advancements are hoping to improve the overall design of the pacemaker. Specifically, testing is being done on leadless pacemakers where the pacemaker requires no leads which would eliminate the need to replace them due to mechanical stress [9]. This intern would also eliminate the risk of leads breaking which leads to further medical complications [9].

An example of this would be the "single component leadless pacemaker" that is continually being tested in humans at both St. Jude Medical (Nanostim) and Medtronic (Micra) [7]. Here, the pulse generator, and sensing and pacing electrodes are integrated into one single capsule that is placed directly into the ventricle [7].

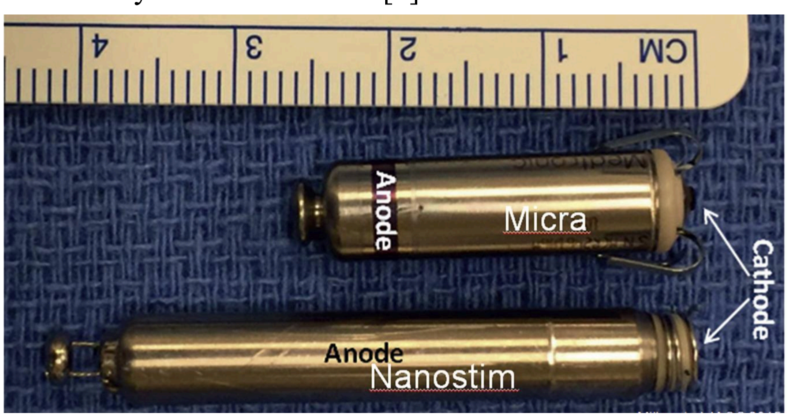


Figure 9. A photo of single component pacemakers, Micra and Nanostim. [7]

Final Conclusions

As closely examined, transmembrane potential of a neuron, action potential of muscle contraction and cardiac action potential work together in conjunction to allow the heart to beat. When any one of these systems malfunction, then the use of a pacemaker is required for the heart to continue beating. Advances in pacemaker technology allow for pacemakers to be less invasive than previous models and are still being tested nowadays.

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